

Assignment: Windows Server

Module -12: Installation, Storage, and Compute with Windows Server

1) What two options are provided in the type of installation window during Windows Server 2016 installation?

Answer:

- 1. Server Core Installation**
- 2. Server with Desktop Experience (GUI)**

2) Write the steps to configure a server step by step

Answer:

- 1. Install Windows Server 2016.**
- 2. Set computer name and administrator password.**
- 3. Configure IP address and DNS settings.**
- 4. Install required server roles and features.**
- 5. Join the server to a domain if needed.**
- 6. Configure firewall and security settings.**
- 7. Update Windows and install drivers.**
- 8. Verify server services and connectivity.**

3) What are the Pre-installation tasks?

Answer:

- Check hardware compatibility and system requirements.**

- **Backup important data.**
- **Prepare bootable installation media.**
- **Configure BIOS/UEFI settings.**
- **Plan disk partitions and network settings.**
- **Ensure power backup and internet connectivity.**

4) What are the Post-installation tasks?

Answer:

- **Install drivers and Windows updates.**
- **Configure IP address and hostname.**
- **Enable firewall and security settings.**
- **Install server roles and features.**
- **Create user accounts and permissions.**
- **Verify backup and monitoring settings.**

5) What is the standard upgrade path for Windows Server?

Answer:

Windows Server can usually be upgraded to the next higher version.

Example:

- **Windows Server 2008 → 2012**
- **Windows Server 2012 → 2016**
- **Windows Server 2016 → 2019**

Direct upgrades depend on edition compatibility and Microsoft support.

6) What is the Physical structure of AD?

Answer:

The physical structure of Active Directory consists of:

- **Domain Controllers**
- **Sites**
- **Subnets**

It defines how network resources are physically connected and replicated.

7) What are the Logical components of Active Directory?

Answer:

Logical components of AD include:

- **Domains**
- **Trees**
- **Forests**
- **Organizational Units (OUs)**
- **Users and Groups**

These components help organize and manage network resources logically.

8) What is the Full form of LDAP?

Answer:

LDAP = Lightweight Directory Access Protocol

It is used to access and manage directory information services over a network.

9) What is the location of the AD database?

Answer:

The Active Directory database file is stored at:

C:\Windows\NTDS\NTDS.dit

This file stores all AD objects and directory information.

10) What is Child DC?

Answer:

A Child Domain Controller belongs to a child domain within a parent domain structure.

It helps manage users and resources for a specific branch or department.

Child domains improve administration and security management.

11) Explain the term Forest in AD

Answer:

A Forest is the top-level structure in Active Directory.

It contains one or more domain trees sharing a common schema and global catalog.

All domains in a forest trust each other automatically.

It provides centralized management and security.

12. What is Active Directory? Check all that apply.

- ☐ **An open-source directory server**
- ☐ **A Windows-only implementation of a directory server**
- ☐ **Microsoft's implementation of a directory server**
- ☐ **An LDAP-compatible directory server**

Answers:

- **Windows-only implementation of a directory server**
- **Microsoft's implementation of a directory server**
- **An LDAP-compatible directory server**

13. When you create an Active Directory domain, what's the name of the default user account?

- ☐ **Superuser**
- ☐ **Root**
- ☐ **Username**
- ☐ **Administrator**

Answer:

- **Administrator**

14. AD domain provides which of the following advantages? Check all that apply.

- ☐ **Centralized authentication**
- ☐ **More detailed logging**
- ☐ **Centralized management with GPOs**
- ☐ **Better performance**

Answers:

- **Centralized authentication**
- **More detailed logging**
- **Centralized management with GPOs**

15) What are the minimum hardware requirements for installing Windows Server 2016?

Answer:

- **Processor: 1.4 GHz 64-bit processor**
- **RAM: 512 MB minimum (2 GB recommended for GUI)**
- **Storage: 32 GB minimum**
- **Network Adapter: Gigabit Ethernet adapter**
- **DVD drive or USB boot support**

16) Explain the different editions of Windows Server 2016 and their features

Answer:

1. Datacenter Edition

- **Supports unlimited virtualization.**
- **Advanced features like Storage Spaces Direct.**

2. Standard Edition

- **Supports limited virtualization.**
- **Suitable for small and medium businesses.**

3. Essentials Edition

- **Designed for small businesses with fewer users.**
- **Simple management and lower cost.**

17) Walk through the steps of installing Windows Server 2016 using GUI mode

Answer:

- 1. Boot system using Windows Server installation media.**
- 2. Select language, time, and keyboard settings.**
- 3. Click “Install Now”.**
- 4. Choose “Windows Server 2016 with Desktop Experience”.**
- 5. Accept license terms.**
- 6. Select Custom Installation.**
- 7. Choose disk partition and click Next.**
- 8. Installation starts and server reboots automatically.**
- 9. Set Administrator password after installation.**

18) Describe the steps for installing Windows Server 2016 in Server Core mode

Answer:

- 1. Boot using installation media.**
- 2. Select “Windows Server 2016 Server Core”.**
- 3. Accept license agreement.**
- 4. Choose installation disk and start installation.**
- 5. After reboot, configure password and network settings using command line tools.**
- 6. Install required roles and features.**

19) How do you configure network settings during Windows Server 2016 installation?

Answer:

- **Open Network and Sharing Center.**
- **Select Ethernet adapter properties.**
- **Configure static IP address, subnet mask, gateway, and DNS server.**
- **Save settings and test network connectivity using ping command.**

20) Explain the process of promoting a Windows Server to a Domain Controller

Answer:

- 1. Install Active Directory Domain Services (AD DS) role.**
- 2. Open Server Manager and select “Promote this server to a domain controller”.**
- 3. Choose new forest or existing domain.**
- 4. Configure DNS and domain settings.**
- 5. Set DSRM password.**
- 6. Complete installation and restart the server.**

21) What is Active Directory Domain Services (AD DS), and what are its key components?

Answer:

AD DS is a Windows service used for centralized management of users, computers, and resources in a network.

Key Components:

- **Domain Controllers**

- **Domains**
- **Trees**
- **Forests**
- **Organizational Units (OUs)**
- **Global Catalog**

22) How do you create a new Active Directory user account in Windows Server?

Answer:

- 1. Open “Active Directory Users and Computers”.**
- 2. Select the Organizational Unit (OU).**
- 3. Right-click → New → User.**
- 4. Enter username and login details.**
- 5. Set password and permissions.**
- 6. Click Finish to create the account.**

23) Explain the process of creating and managing Group Policy Objects (GPOs) in Windows Server 2016 or 2019

Answer:

- 1. Open Group Policy Management Console (GPMC).**
- 2. Right-click the domain or OU and select “Create a GPO”.**
- 3. Edit policy settings like security, software, or user restrictions.**
- 4. Link the GPO to users or computers.**
- 5. Apply changes using gpupdate /force**

24) What are Organizational Units (OUs) in Active Directory, and how do you use them?

Answer:

Organizational Units (OUs) are containers used to organize users, groups, and computers within Active Directory.

They help simplify administration and policy management.

Administrators can apply specific Group Policies to different OUs.

Example: Separate OUs for HR, IT, and Sales departments.

25) Describe the process of delegating administrative privileges in Active Directory

Answer:

- 1. Open Active Directory Users and Computers.**
- 2. Right-click the OU and select “Delegate Control”.**
- 3. Add users or groups to delegate permissions.**
- 4. Choose tasks like resetting passwords or managing users.**
- 5. Finish the wizard to apply delegated permissions.**